Advanced Biotechnology

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UNIVERSITY OF CAMBRIDGE

Lecture 10

Bioprocessing:
Cell removal (part II)
+
Cell disruption (part I)



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Lecture outline

1) Principles of centrifugation for cell removal and centrifuge designs
2) Centrifugation theory
3) Overview of cell disruption
4) Cell disruption methods: Mechanical methods

Chapter 11 Bioprocess Engineering Principles (11.4 and 11.5)

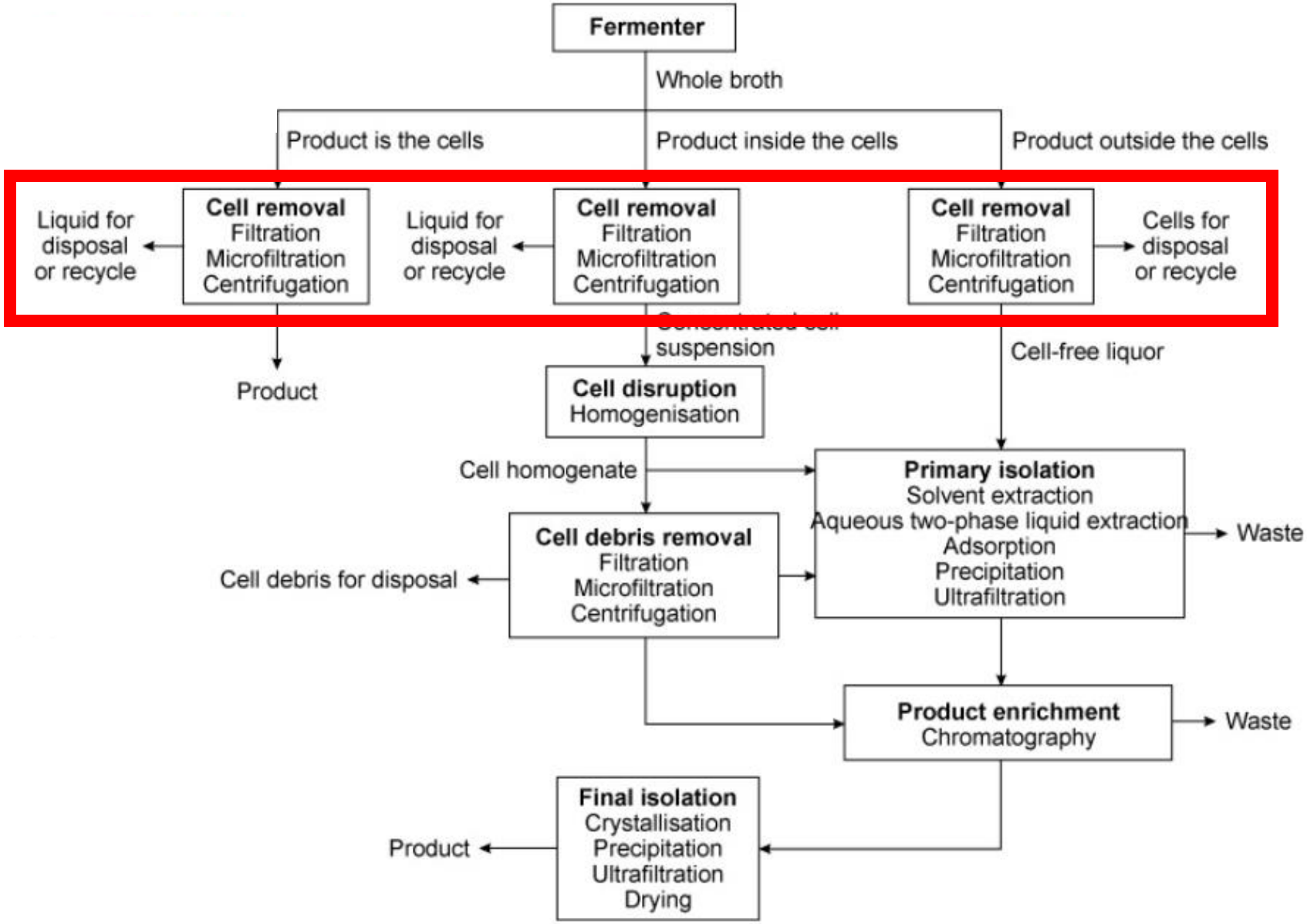
Learning outcomes

1) Discuss basic design of tubular bowl and disc-stack bowl centrifuge
2) Mathematically describe centrifugation principles (Σ theory, volumetric flow rate)
3) Discuss purpose of cell disruption in the context downstream processing
4) Discuss principles of mechanical methods for cell disruption
5) Mathematically estimate protein extraction efficiency for mechanical cell disruption methods

Lecture outline

1) **Principles of centrifugation for cell removal and centrifuge designs**
2) Centrifugation theory
3) Overview of cell disruption
4) Cell disruption methods: Mechanical methods

Downstream process logic



Centrifugation

- Like filtration, separates solids from liquid medium
- Works by rotating the sample so to produce high centrifugal acceleration, which causes particles to sediment out of the liquid phase
- Most effective
  - Large particles
  - Low liquid viscosity
  - Large density difference between the particles and liquid

Centrifugation

- Conditions may not be easy to establish with cell cultures (cells may not be very dense, media can be dense, bacterial cells are small)

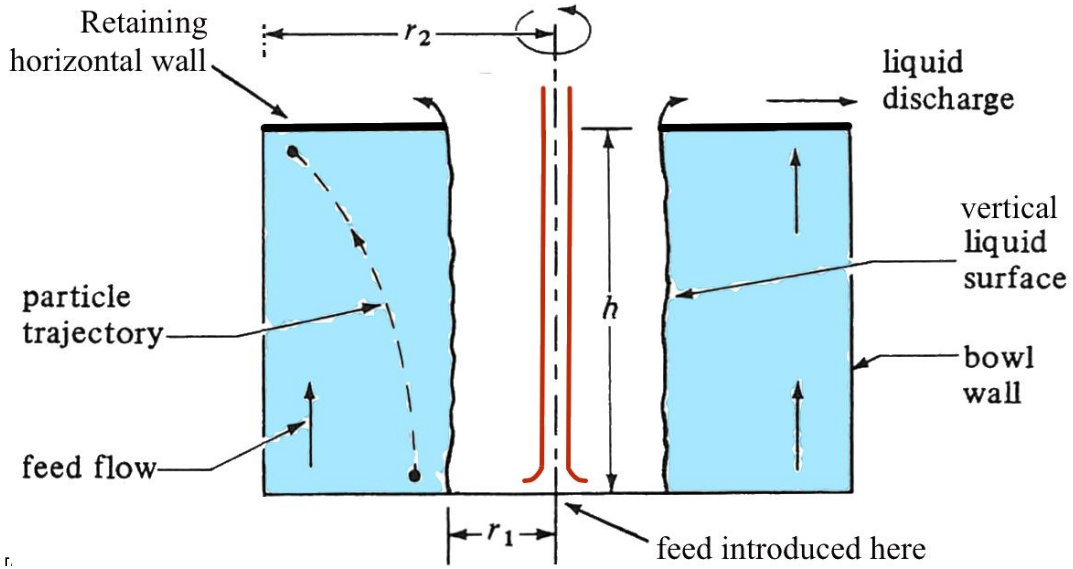
- Limitations are overcome by designing large centrifuges that operate at high rotational speeds, although physical and safety limitations can limit size and operation

Types of centrifuge

- Centrifuges can be classified on the design of their internal structure/geometry

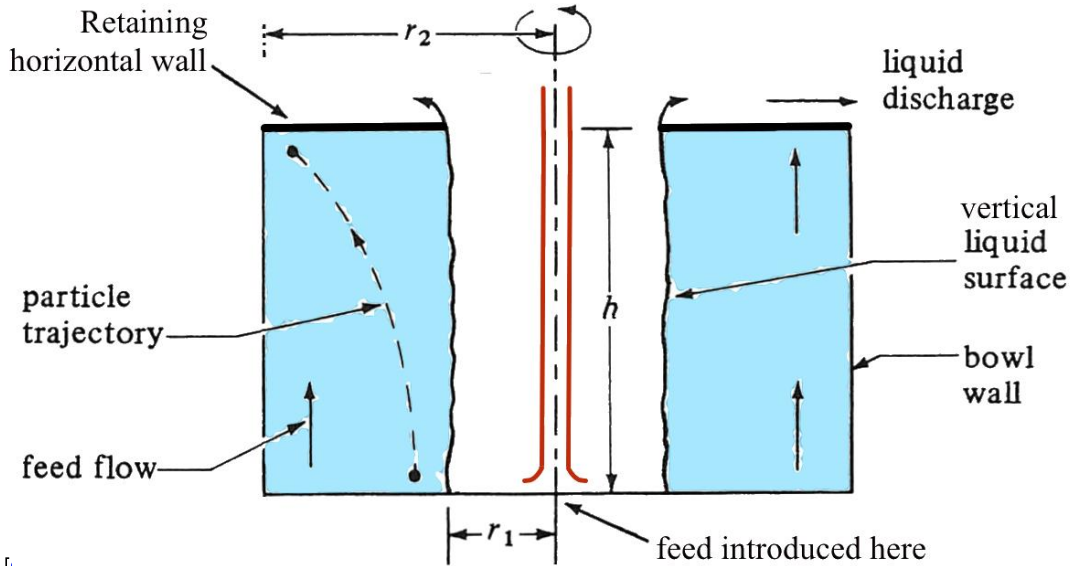
- We will discuss two types:
  1. Tubular bowl centrifuge
  2. Disc stack bowl centrifuge

Tubular bowl centrifuge



- Commonly used in food and bio-pharma
- The equipment consists of a rotating tube and a nozzle at the base of the unit where feed is introduced
- As particles flow upwards, they are spun outwards, colliding with the walls of the bowl
- The liquid continues to flow upwards, leaving via a weir at the top of the bowl

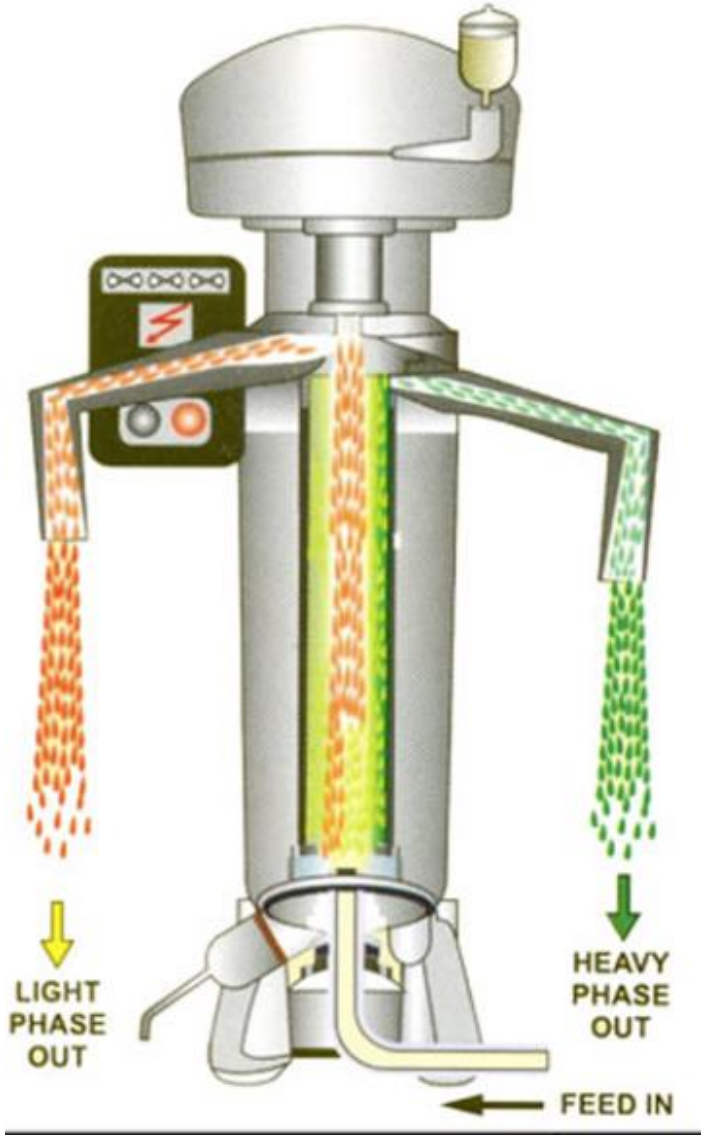
Tubular bowl centrifuge



- The time that it takes from the liquid to exit the top of the tube is the **residence time**, which **decreases with increasing flow rate**
- Over a **sedimentation time**, centrifugal force will cause the particles to migrate towards the wall of the tube
- The sedimentation time is faster for greater rotations speed (larger/denser particles, lower viscosity…)
- Particles are efficiently separated if **the sedimentation time is shorter than the residence time**

Tubular bowl centrifuge

- **Batch centrifugation** is common for tubular bowl centrifuge: the solids need to be removed once they fill the tube up to the escape point of the liquid
- Some models an operate as **continuous process** (scheme on the right), using a heavy-phase outlet close to the outer wall

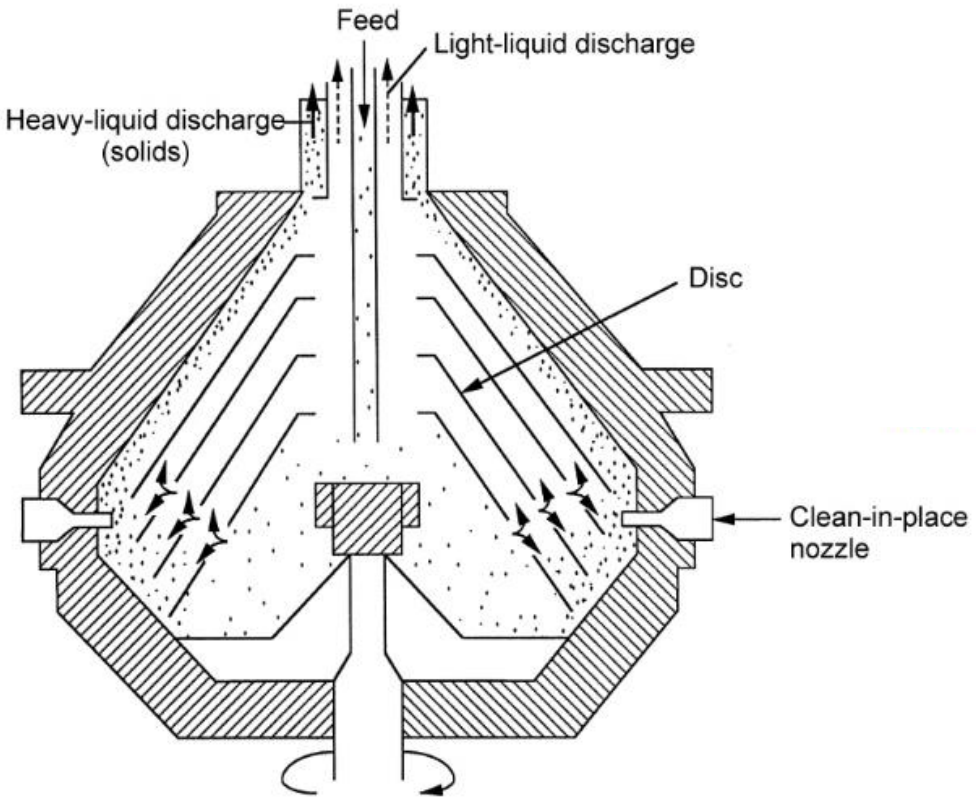


Tubular bowl centrifuge

- Useful for difficult separations that require high centrifugal forces – up to 16000 g
- Foaming can be a problem
- Easy to clean



Disc-stack centrifuge

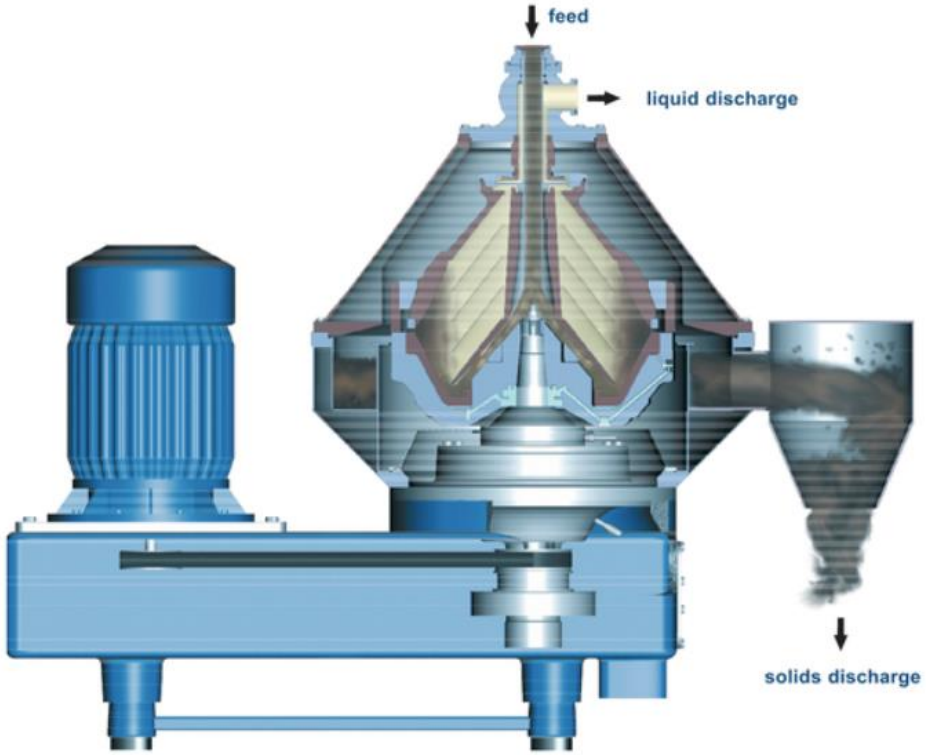


- Centrifuge contain conical shear sheets of metal that are stacked on top of each other with clearances of approximately 0.3 mm

  As the bowl rotates, the discs split liquid into layers

  The denser liquid containing the solids is displaced outwards whereas lighter liquid remains close to the centre
- Both streams are withdrawn from the centrifuge as shown in the figure on the left
- Solids can be removed periodically from nozzles (batch mode) or continuously from the heavy-liquid discharge outlet (continuous mode, more common)

Disc-stack centrifuge



- Foam can be controlled
- Difficult to clean compared to tube bowl
- Easier to use in continuous mode compared with tubular bowl

Disc-stack Vs tubular

Centrifuge type	Bowl diameter (cm)	Speed (rpm)	Maximum centrifugal force (×g)	Liquid throughput (l min^{-1})	Motor size (kW)
Tubular bowl	10	15,000	13,200	0.4-40	1.5
	13	15,000	15,900	0.8-80	2.2
Disc stack	18	12,000	14,300	0.4-40	0.25
	33	7500	10,400	20-200	4.5
	61	4000	5500	80-800	5.6

Centrifugation vs filtration

Centrifugation

Pros:

- Easier to make sterile
- No consumables cost
- Easier to run in continuous mode

Cons:

- Higher capital cost
- Many moving parts/seals that can fail
- Safety

Filtration

Pros:

- Lower capital costs
- Easier to separate small cells

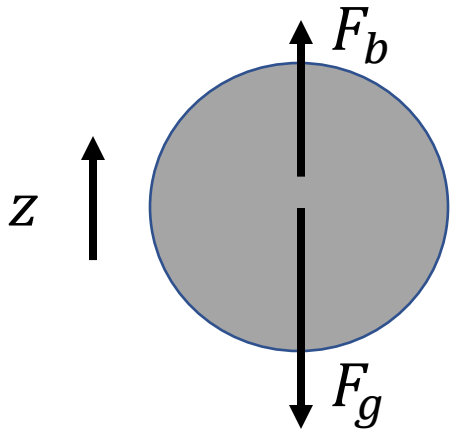
Cons:

- Nonsterile
- Consumables costs (filters)

Lecture outline

1) Principles of centrifugation for cell removal and centrifuge designs
2) **Centrifugation theory**
3) Overview of cell disruption
4) Cell disruption methods: Mechanical, Non-mechanical physical, Chemical and Biological methods

Centrifugation theory: sedimentation



A **spherical particle** in a static fluid experiences two forces:

1) Gravitational force:

$$F_g = -m_p\, g = -V_p\, \rho_p\, g$$

Where m_p, V_p, and ρ_p are, respectively, the mass, volume, and density of the particle.

2) Buoyancy force:

$$F_b = V_p\, \rho_l\, g$$

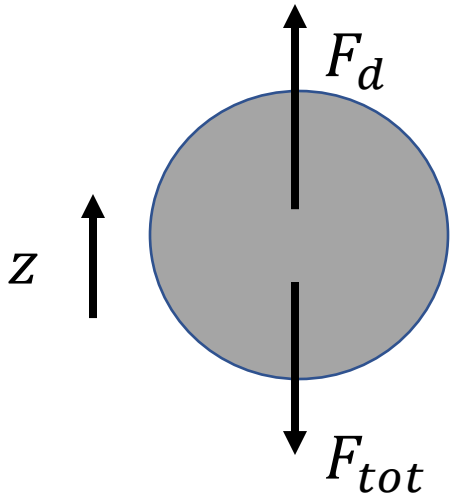
Where ρ_p is the density of the liquid (note different sign).

The net force acting on the particle is

$$F_s = F_g + F_b = -V_p(\rho_p - \rho_l)g$$

If $\rho_p > \rho_l$, the particle will sediment

Centrifugation theory: sedimentation



A **spherical particle** moving with a velocity u in a Newtonian fluid at low Reynolds number experiences a **frictional force** equal to

$$F_d = 3\pi\,\mu\,D_p\,u$$

Where D_p is the particle diameter and μ the viscosity of the liquid

At low Re **inertial effects are negligible**, so $F_s = -F_d$ (the net force acting on the particle is always zero)

$$3\pi\,\mu\,D_p\,u_g = V_p(\,\rho_p - \rho_l)g = \frac{\pi}{6}D_p^3(\,\rho_p - \rho_l)g$$

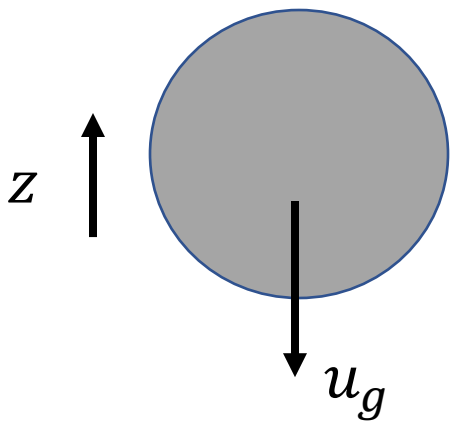
Where we used $V_p = \frac{\pi}{6}D_p^3$. The (terminal) sedimentation velocity u_g is thus

$$u_g = \frac{D_p^2(\,\rho_p - \rho_l)g}{18\,\mu}$$

This is the **Stokes Law** for the sedimentation velocity of the particle in the gravitational field.

Centrifugation theory: sedimentation

$$u_g = \frac{D_p^2(\rho_p - \rho_l)g}{18\,\mu}$$



Considering a (spherical) bacterium with density $\rho_p = 1.3 \times 10^3\ \mathrm{kg\,m^{-3}}$ in water ($\rho_l = 10^3\ \mathrm{kg\,m^{-3}}$, $\mu = 10^{-3}\ \mathrm{Pa\,s^{-1}}$), and $D_p = 1\ \mu\mathrm{m}$, we get

$$u_g = 1.6 \times 10^{-8}\ \mathrm{m\,s^{-1}} = 0.16\ \mu\mathrm{m\,s^{-1}}$$

This is very slow. The bacteria would take ~70 days to sediment by 0.1m!

Centrifugation theory: centrifugal force

In a centrifuge, the fluid rotates along with the bowl with angular velocity ω. A particle at distance r from the rotation axis will thus experience a centrifugal acceleration

$$a_c = r\omega^2$$

We can calculate the sedimentation velocity under the centrifugal rather than gravitational acceleration by simply replacing g with a_c

$$u_c = \frac{D_p^2(\rho_p - \rho_l) r\omega^2}{18\,\mu}$$

Centrifugation theory: centrifugal force

The **G-factor of a centrifuge** is simply the ratio between a_c and g.

For a typical centrifuge with $\mathrm{G} = 10{,}000$, we can re-calculate the sedimentation velocity for the bacterium in the previous example

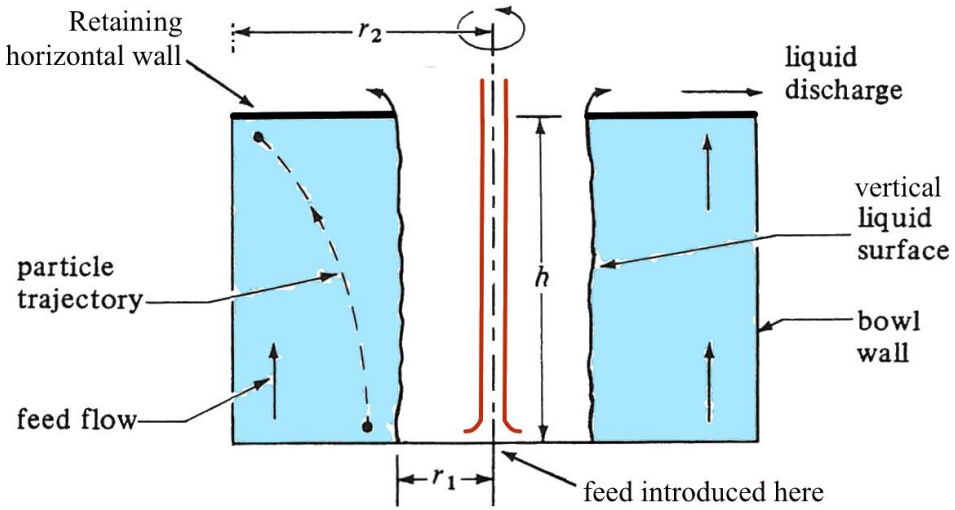
$$u_c = \frac{D_p^2(\rho_p - \rho_l) G g}{18\,\mu} = \frac{D_p^2(\rho_p - \rho_l) r \omega^2}{18\,\mu}$$

$\rho_p = 1.3 \times 10^3\ \mathrm{kg\,m^{-3}}$, $\rho_l = 10^3\ \mathrm{kg\,m^{-3}}$, $\mu = 10^{-3}\ \mathrm{Pa\,s^{-1}}$, $D_p = 1\ \mu\mathrm{m}$

$$u_c = 1.6 \times 10^{-4}\ \mathrm{m\,s^{-1}} = 160\ \ \mu\mathrm{m\,s^{-1}}$$

About 10 minutes to sediment over 0.1m. Much better than 70 days.

Centrifugation theory: throughput of a tubular bowl centrifuge

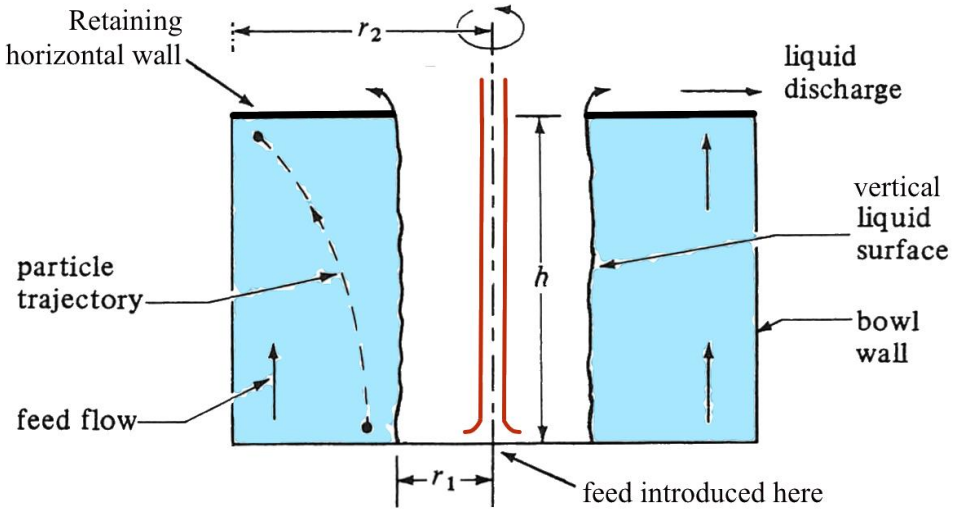


- For the particle to be "captured" by the centrifuge, we require that it hits the outer wall before the feed flow pushes it out of the tube.
- **Sedimentation time < residence time**
- For a tubular bowl centrifuge, t_{cut} is the time needed for a particle to sediment from $\frac{r_1+r_2}{2}$ to r_2 under centrifugal acceleration $r\omega^2$ (compute this at home!)

$$t_{cut} = \frac{18\,\mu}{D_p^2(\rho_p - \rho_l)\omega^2} \ln\left(\frac{2\,r_2}{r_1 + r_2}\right)$$

- We can use t_{cut} **to approximate the sedimentation time**

Centrifugation theory: centrifuge throughput of a tubular bowl centrifuge



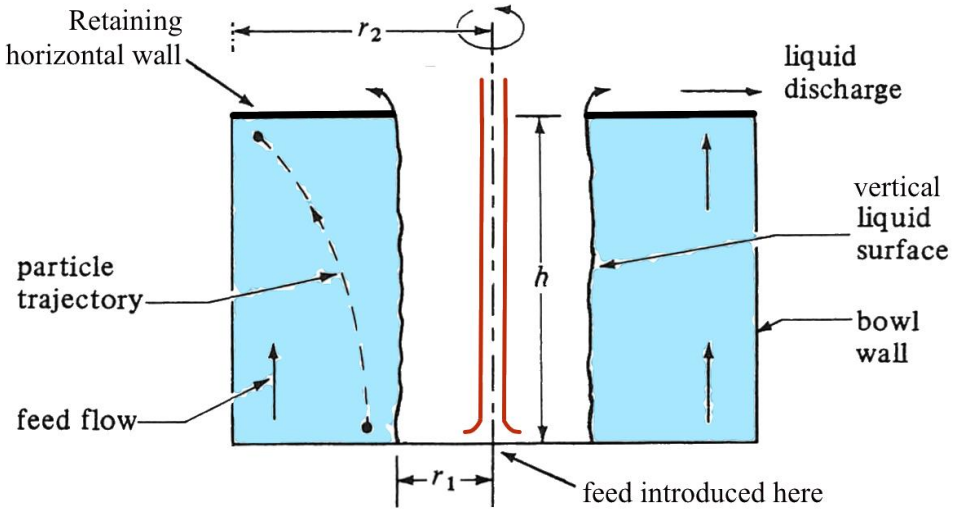
The **residence time** in the centrifuge is

$$t_{res} = \frac{V_c}{Q}$$

where

- Q $(\mathrm{m}^3\ \mathrm{s}^{-1})$ is the volumetric flow rate of the centrifuge
- $V_c = \pi(r_2^2 - r_1^2)h$ is the volume of the centrifuge
- h is the height of the tube

Centrifugation theory: centrifuge throughput of a tubular bowl centrifuge



Cells are separated if the residence time is smaller than the sedimentation time, $t_{res} > t_{cut}$, and hence if the volumetric flow rate is

$$Q < \frac{V_c}{t_{cut}} = Q_{cut}$$

Where

$$Q_{cut} = \frac{V_c}{t_{cut}} = \frac{D_p^2(\rho_p - \rho_l)\omega^2}{18\,\mu \ln\left(\frac{2\,r_2}{r_1 + r_2}\right)} \pi(r_2^2 - r_1^2)h$$

is the maximum volumetric flow-rate allowed by the tubular bowl centrifuge

Example 1

A lab-scale tubular centrifuge has liquid surface radius $r_1 = 16.5\ \mathrm{mm}$ and outer radius $r_2 = 22.2\ \mathrm{mm}$, respectively, bowl height $\mathrm{h} = 115\ \mathrm{mm}$ and angular speed $\omega = 10{,}000\ \mathrm{rpm}$.

It is used to separate bacteria with density $\rho_p = 1.340 \times 10^3\ \mathrm{kg\,m^{-3}}$ and diameter $D_p = 0.7\ \mu\mathrm{m}$ from broth with $\rho_l = 1.010 \times 10^3\ \mathrm{kg\,m^{-3}}$ and $\mu = 10^{-3}\ \mathrm{Pa\ s^{-1}}$.

Calculate:

a) the centrifugal acceleration experienced by the bacteria at $r = r_2$ (in unit of g)

b) Q_{cut}

Solution: a

Compute centrifugal acceleration at r_2

$$a_c(r_2) = r_2\omega^2 = 2.22 \times 10^{-2}\ \mathrm{m} \times \left(2\pi \times \frac{10{,}000}{60}\ \mathrm{rad\ s^{-1}}\right)^2 = 24385\ \mathrm{m\ s^{-2}} = 2482\ g$$

Note that we converted ω in $\mathrm{rad\ s^{-1}}$

$$\omega = 1047\ \mathrm{rad\ s^{-1}}$$

Solution: b

Compute more realistic cutoff volumetric flow rate Q_{cut}

$$Q_{cut} = \frac{V_c}{t_{cut}} = \frac{D_p^2(\rho_p - \rho_l)\omega^2}{18\,\mu \ln\left(\frac{2\,r_2}{r_1 + r_2}\right)}\,\pi(r_2^2 - r_1^2)h$$

$$Q_{cut} = \frac{(0.7 \times 10^{-6}\ \mathrm{m})^2(330\ \mathrm{kg\ m^{-3}})(1047\ \mathrm{rad\ s^{-1}})^2}{18 \times 10^{-3}\ \mathrm{Pa\ s^{-1}} \ln\left(\frac{2 \times 0.022\mathrm{m}}{0.022\mathrm{m} + 0.016\mathrm{m}}\right)}\,\pi\left((0.022\ \mathrm{m})^2 - (0.0165\ \mathrm{m})^2\right) \times 0.115\mathrm{m}$$

$$Q_{cut} = 5.55 \times 10^{-6}\ \mathrm{m^3 s^{-1}} = 5.55\ \mathrm{ml\ s^{-1}}$$

Centrifugation Sigma theory

Consider the cutoff volumetric flow rate

$$Q_{cut} = \frac{D_p^2(\rho_p - \rho_l)\omega^2}{18\,\mu \ln\left(\frac{2\,r_2}{r_1 + r_2}\right)}\,\pi(r_2^2 - r_1^2)h$$

We multiply and divide Q_{cut} by the gravitational sedimentation velocity

$$u_g = \frac{D_p^2(\rho_p - \rho_l)g}{18\,\mu}$$

$$Q_{cut} = \frac{D_p^2(\rho_p - \rho_l)\omega^2}{18\,\mu \ln\left(\frac{2\,r_2}{r_1 + r_2}\right)}\,\pi(r_2^2 - r_1^2)h\,\frac{u_g}{\frac{D_p^2(\rho_p - \rho_l)g}{18\,\mu}}$$

Centrifugation Sigma theory

$$Q_{cut} = \frac{\pi(r_2^2 - r_1^2)h\ \omega^2}{g \ln\left(\frac{2\ r_2}{r_1 + r_2}\right)} u_g = 2\ \Sigma\ u_g$$

Where

$$\Sigma(r_1, r_2, h, \omega) = \frac{\pi(r_2^2 - r_1^2)h\ \omega^2}{2\ g \ln\left(\frac{2\ r_2}{r_1 + r_2}\right)}$$

- Σ **is only a function of the characteristics of the centrifuge**, not of the sample
- Σ has the dimensions of an area, and is physically interpreted as the **equivalent surface area to achieve a volumetric sedimentation throughput Q_{cut} by gravity only**

Centrifugation Sigma theory

Because it is independent on the characteristics of the broth, Σ is useful to **compare two centrifuges to be used on the same sample**, e.g. for scaling up

$$Q_1 = 2\,\Sigma_1 u_g$$
$$Q_2 = 2\,\Sigma_2 u_g$$

Dividing, we obtain (because u_g is a characteristic of the sample)

$$\frac{Q_1}{Q_2} = \frac{\Sigma_1}{\Sigma_2}$$

Therefore, If one would like to increase the throughput of a centrifuge by a factor X, also Σ needs to increase by the same factor

Different formulas for Sigma can be found…

For the tubular bowl centrifuge, we derived:

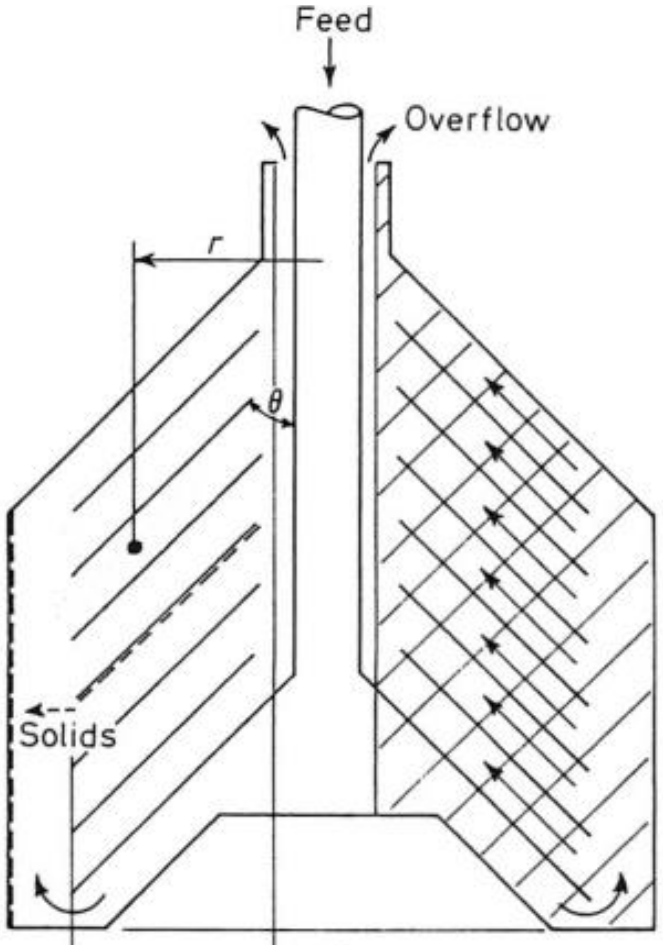
$$\Sigma(r_1, r_2, h, \omega) = \frac{\pi(r_2^2 - r_1^2) h\ \omega^2}{2g \ln\left(\frac{2\ r_2}{r_1 + r_2}\right)}$$

Which however depends on the choice of the cutoff time, which is somewhat arbitrary

Other expressions are also used commonly, for instance

$$\Sigma(r_1, r_2, h, \omega) = \frac{\pi\ \omega^2\ h}{2\ g}(3r_2^2 + r_1^2)$$

Centrifugation Sigma theory



Equations for Σ depend on the centrifuge design. For a disc stack centrifuge,

$$\Sigma = \frac{2\pi\omega^2(N-1)}{3g\tan\theta}(r_2^3 - r_1^3)$$

where N is the number of discs in the stack; r_2 is larger radius of the disc; r_1 is the smaller radius of the disc and θ is the half cone angle

Example 2:

Given the parameters in Example 1:

a) Calculate Σ (using the expression we derived)

b) A larger centrifuge has double the physical dimensions of the one in Example 1 ($r_1 = 33.0\ \mathrm{mm}, r_2 = 44.4\ \mathrm{mm}, \mathrm{h} = 230\ \mathrm{mm}$). What angular speed is needed for the larger centrifuge to match the throughput of the smaller?

Example 2:

Parameters in Example 1:

$$r_1 = 16.5\ \text{mm}$$
$$r_2 = 22.2\ \text{mm}$$
$$\text{h} = 115\ \text{mm}$$
$$\omega = 10{,}000\ \text{rpm}$$
$$\rho_p = 1.340 \times 10^3\ \text{kg m}^{-3}$$
$$D_p = 0.7\ \mu\text{m}$$
$$\rho_l = 1.010 \times 10^3\ \text{kg m}^{-3}$$
$$\mu = 10^{-3}\ \text{Pa s}^{-1}$$

Solution: a

Calculate Σ

$$\Sigma(r_1, r_2, h, \omega) = \frac{\pi(r_2^2 - r_1^2) h\ \omega^2}{g \ln\left(\frac{2\, r_2}{r_1 + r_2}\right)}$$

$$\Sigma = \frac{\pi\left((0.022\ \text{m})^2 - (0.0165\ \text{m})^2\right) \times 0.115\text{m} \times (1047\ \text{rad s}^{-1})^2}{9.81\ \text{m s}^{-2} \ln\left(\frac{2 \times 0.0222\ \text{m}}{0.0222\ \text{m} + 0.0165\ \text{m}}\right)} = 64.8\ \text{m}^2$$

Solution: b

Compute ω for the larger centrifuge. We note

$$\Sigma(r_1, r_2, h, \omega) = \frac{\pi(r_2^2 - r_1^2)h}{g \ln\left(\frac{2\,r_2}{r_1 + r_2}\right)}\,\omega^2 = f(r_1, r_2, h)\,\omega^2$$

$$f = \frac{\pi(r_2^2 - r_1^2)h}{g \ln\left(\frac{2\,r_2}{r_1 + r_2}\right)}$$

Comparing the two centrifuges

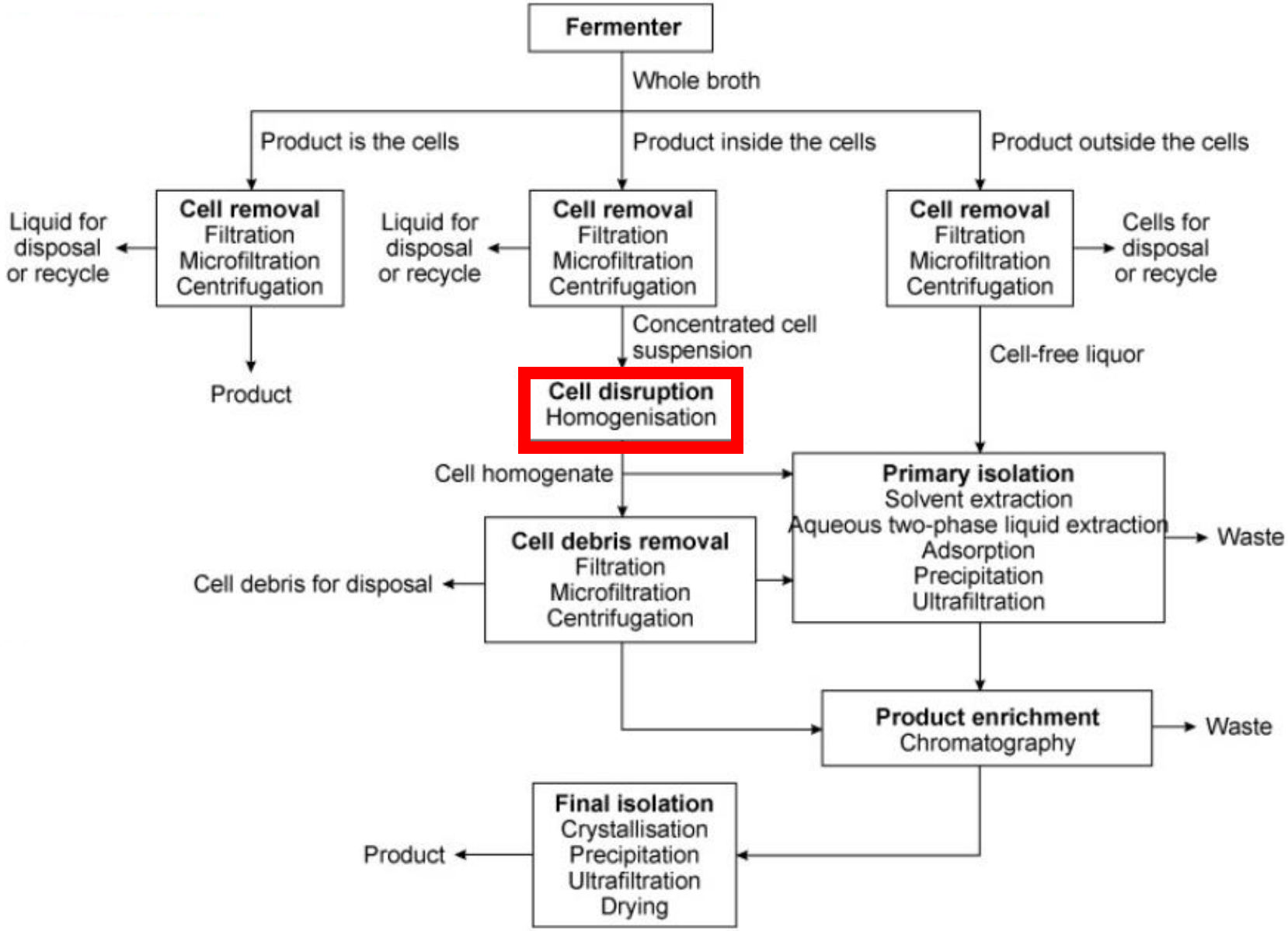
$$\frac{\Sigma_2}{\Sigma_1} = \frac{f_2 \omega_2^2}{f_1 \omega_1^2} = \frac{Q_2}{Q_1} = 1$$

$$\omega_2 = \omega_1 \sqrt{\frac{f_1}{f_2}} = \omega_1 \sqrt{\frac{r_{2,1}^2 - r_{1,1}^2}{r_{2,2}^2 - r_{1,2}^2}} = \frac{\omega_1}{2} = 5{,}000 \text{ rpm}$$

Lecture outline

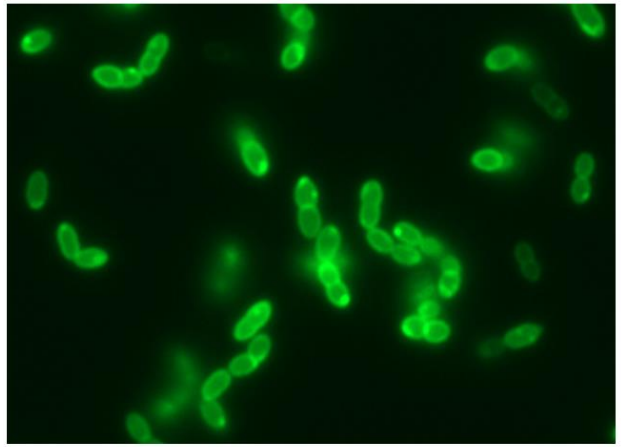
1) Principles of centrifugation for cell removal and centrifuge designs
2) Centrifugation theory
3) **Overview of cell disruption**
4) Cell disruption methods: Mechanical methods

Downstream process logic

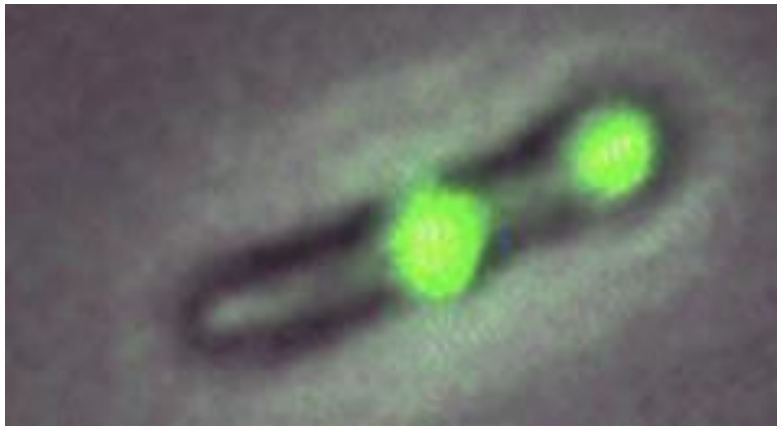


Intra-cellular products extraction requires cell disruption

- Many products we are interested in are intra-cellular (*e.g.* many proteins)
- They require cells to be disrupted before further isolation/purification steps



Lactococcus lactis (found in cheese and yogurt) genetically engineered to express fluorescent proteins



Fluorescent proteins expressed in modified *E. coli*. proteins accumulate in "inclusion bodies" at the poles of the cells.

Cell-disruption strategies

There are many approaches to cell disruption

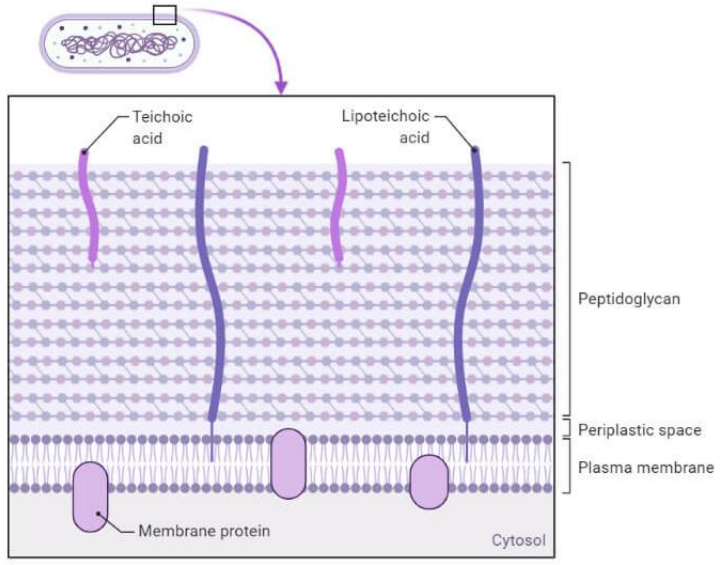
- **Mechanical methods (this lecture)**
  - Bead mill
  - Ultrasonication
  - French press
  - High pressure homogenizer
- **Non-mechanical physical methods (next lecture)**
  - Thermolysis
  - Decompression
  - Osmotic shock
- **Chemical methods (next lecture)**
  - Detergents
  - Solvents
- **Biological methods (next lecture)**
  - Enzymatic digestion

Cell-disruption strategies

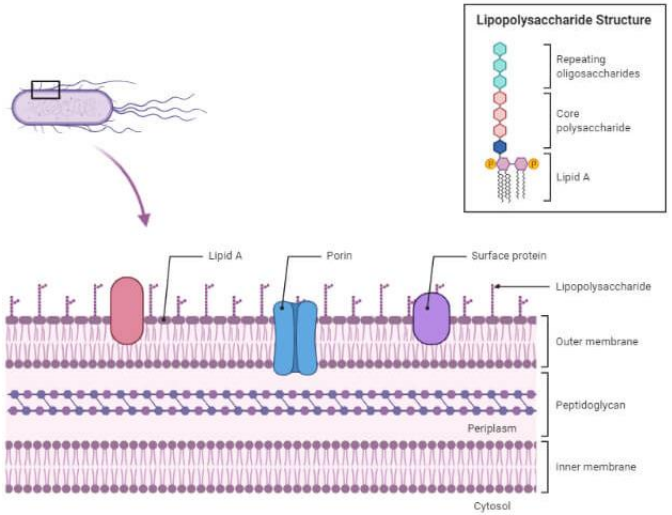
- Some methods are only applicable at **small- or lab-scale**, others scalable
- **Different methods** may be better suited for **different types of cells**
- Different cells are **intrinsically easier or more challenging to disrupt**
- *E.g.* bacteria and yeast (in particular) have tough cell walls difficult to disrupt. Bacteria are small, meaning that they are more difficult to disrupt with mechanical means only
- Mammalian cell lines only have a plasma membrane and no cell wall, so they are easily disrupted

Cell walls

- Microorganisms used in bioprocessing are often surrounded by tough **cell walls**
- **Gram + bacteria** (*e.g. L. lactis*) have a thick peptidoglycan wall. **Gram – bacteria** (*e.g. E. coli*) have a thinner peptidoglycan wall, but it is enveloped by an outer membrane rich in lipopolysaccharides (LPS)



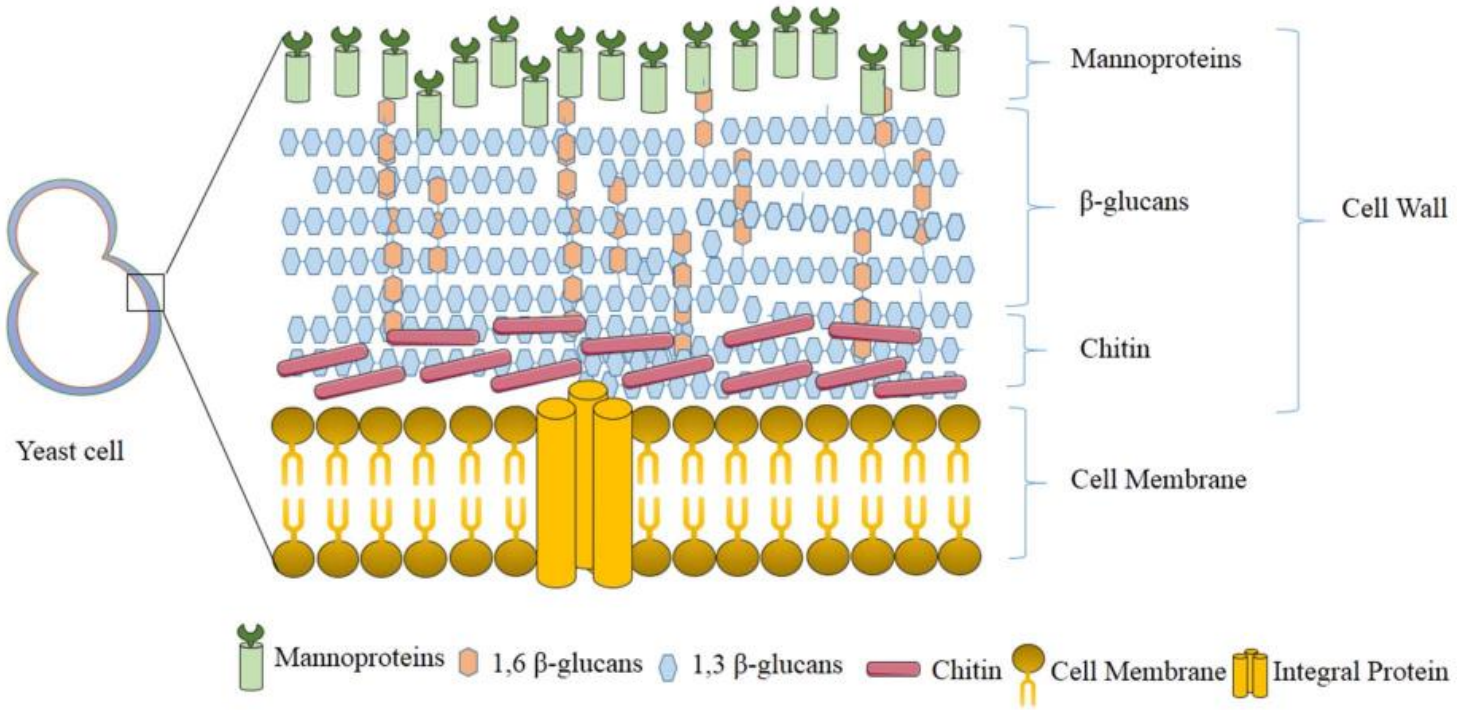
Gram-Positive Bacteria Cell Wall Structure



Gram-Negative Bacteria Cell Wall Structure

Cell walls

- Yeast (*e.g. S. cervisiae*) has a **thick wall with multiple layers**, including polysaccharides, and chitin. It makes yeast very difficult to disrupt



Saccharomyces cervisiae cell wall. World's Poultry Science Journal, 73, 2017

Lecture outline

1) Principles of centrifugation for cell removal and centrifuge designs
2) Centrifugation theory
3) Overview of cell disruption
4) **Cell disruption methods: Mechanical methods**

Mechanical methods

- Include
  - **Bead mill**
  - **Ultrasonication**
  - **French press**
  - **High pressure homogenizer**

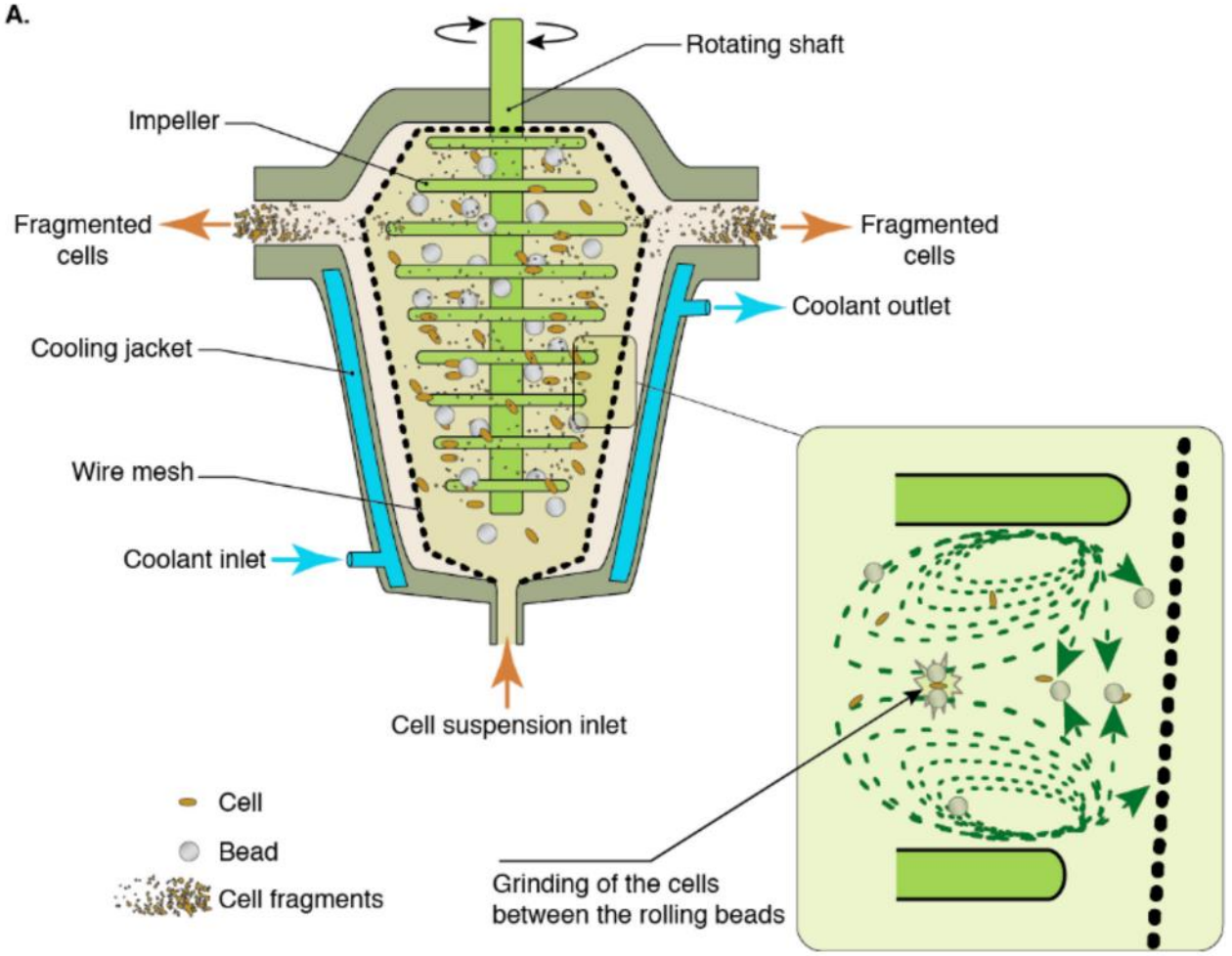
- Cells subjected to high stress via **pressure**, **shearing**, **abrasion**

- Produce substantial heat that can damage compounds (*e.g.* proteins). Often **needs cooling**

- Some methods (*e.g.* French press) only suitable at lab scale. Other methods (high-pressure homogenizer and bead mill) can be used at industrial scale

Mechanical methods: Bead mill

- Chamber with **impellers** (agitators) **rotating** at 1500-2250 rpm
- Filled with **beads**, typically glass
- Bead size: 0.2-1.0 mm
- Bead packing density up to 80% of free volume
- Cell concentration 30-60% volume wet solids



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Mechanical methods: bead mill

Extraction efficiency

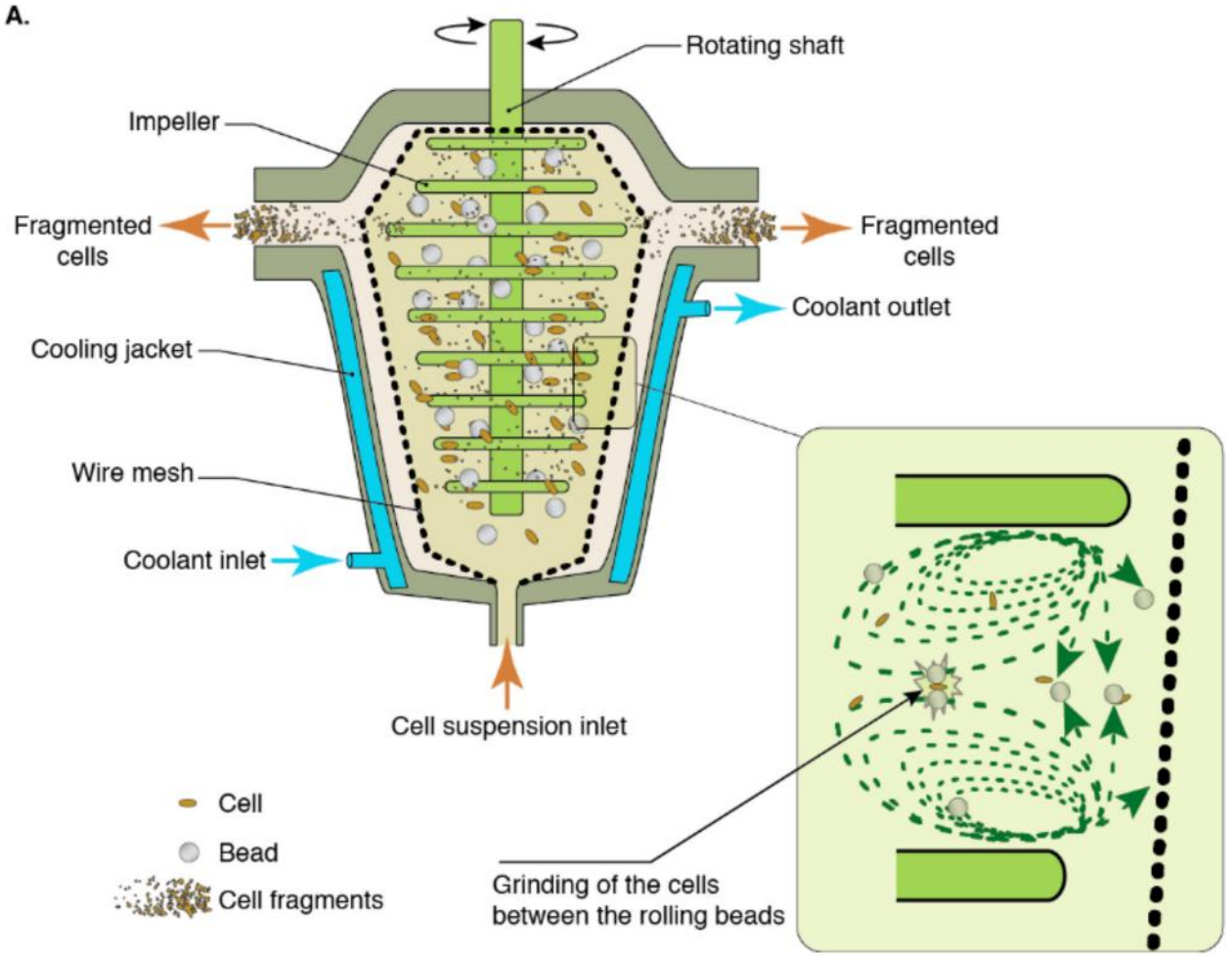
$$\ln\left(\frac{R_{max}}{R_{max}-R}\right) = kt$$

R = protein release in kg of protein per kg of biomass

R_{max} = maximum R achievable

t = processing time

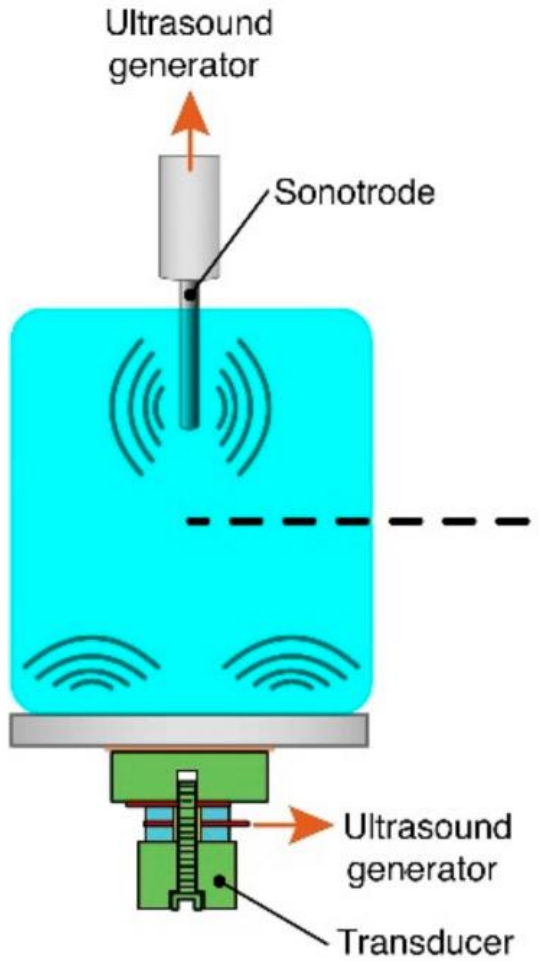
k = constant that depends on process parameters (bead size, bead packing density, rotation speed, temperature)



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Mechanical methods: Ultrasonication

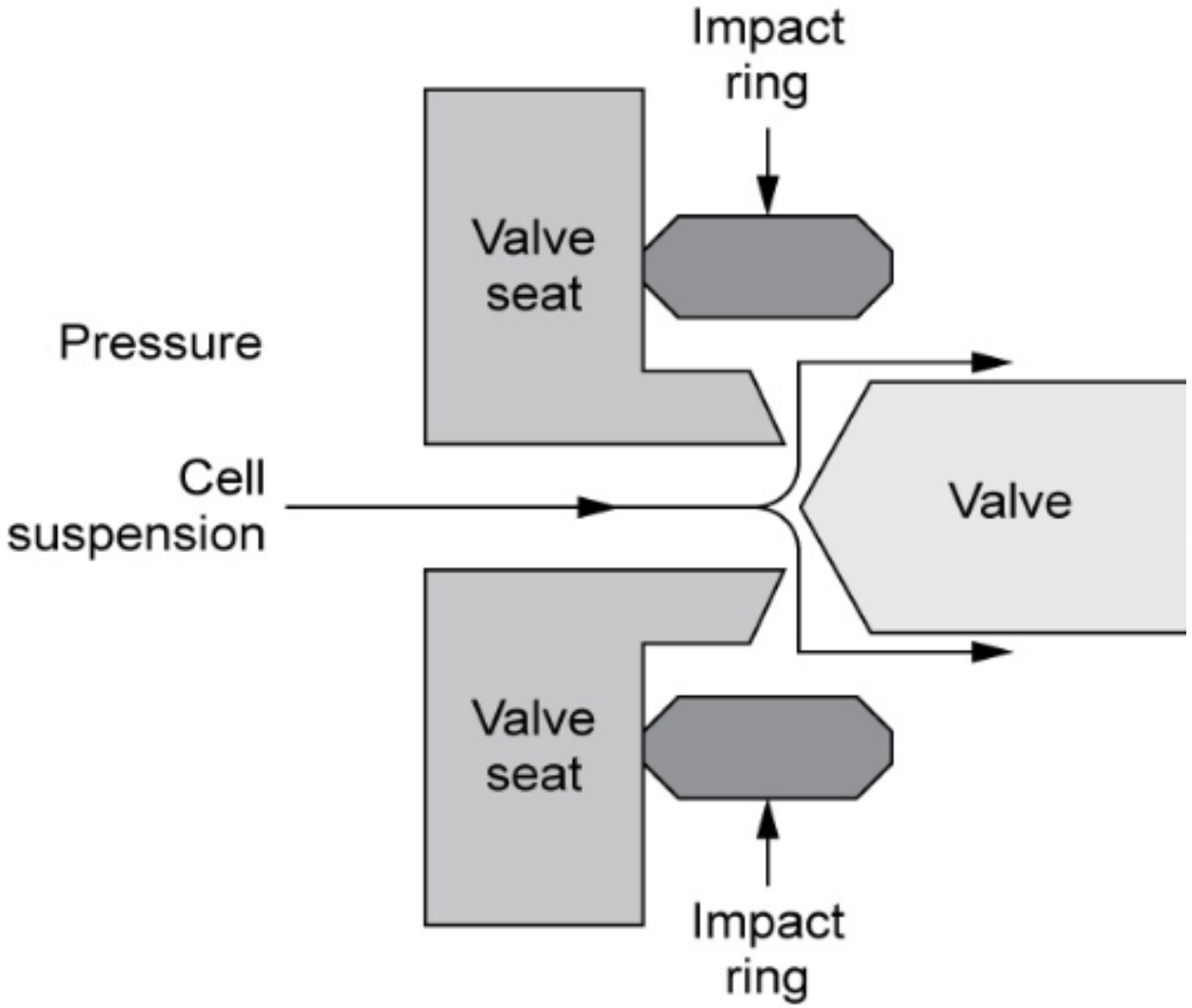
- Ultrasonic vibrations (~20 kHz) are produced by an immersion probe or a vibrating plate
- High energy pressure waves produce **cavitation bubbles**, whose collapse can disrupt cells
- Bacteria can be disrupted in 30-60s, yeast in 2-10 minutes
- Valuable at **small/lab scale**, but difficult to scale-up due to high energy demand and safety concerns (exposure to ultrasounds)



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Mechanical methods: High-pressure homogeniser

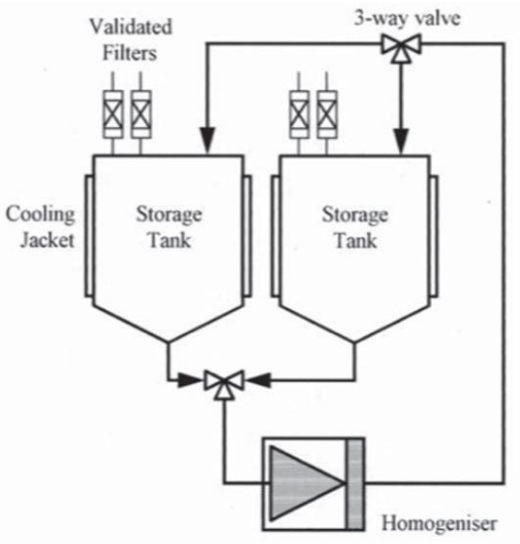
- Cells are put under **high pressure** (1500-2500 bar)
- Pressure is suddenly released by a valve opening forcing the cells to flow at high speed through a small gap
- Cells collide with an **impact ring**, causing disruption
- Can be used at lab and **industrial scale**
- Semi-continuous process



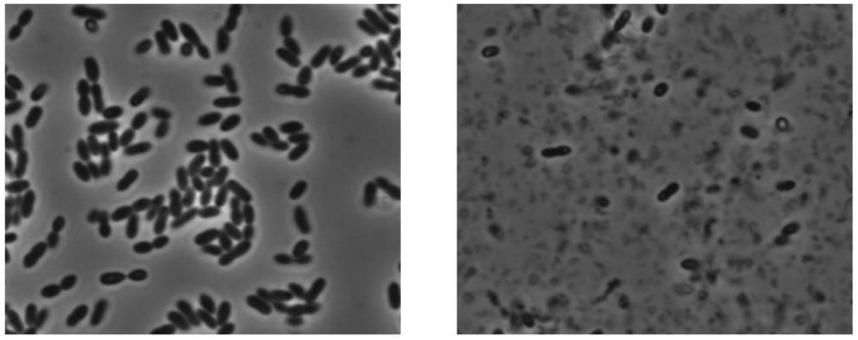
Gaulin-type homogeniser

Mechanical methods: High-pressure homogeniser

- May require **multiple passes** to achieve desired degree of disruption
- This can be achieved with **multi-tank setups**, like the one on the right used for yeast (feed is drawn from one tank, homogenized, put into the other etc.)
- Some cells, e.g. *L. lactis* are very tough (ovoid, gram positive), thus require multiple passes



"Discrete-pass" strategy employed i.e. homogeniser feed is drawn from one tank, and homogenate passed to the other



L. lactis: increasingly popular host for heterologous protein expression
Gram +ve (and ovoid) – so much tougher to break than *E. coli* (Gram –ve)
Several passes in homogeniser required to release protein.

Mechanical methods: High-pressure homogeniser

Extraction efficiency

$$\ln\left(\frac{R_{max}}{R_{max}-R}\right) = kNp^{\alpha}$$

R = protein release in kg of protein per kg of biomass

R_{max} = maximum R achievable

N = number of steps

k = constant that depends on cell type and temperature

p = applied pressure

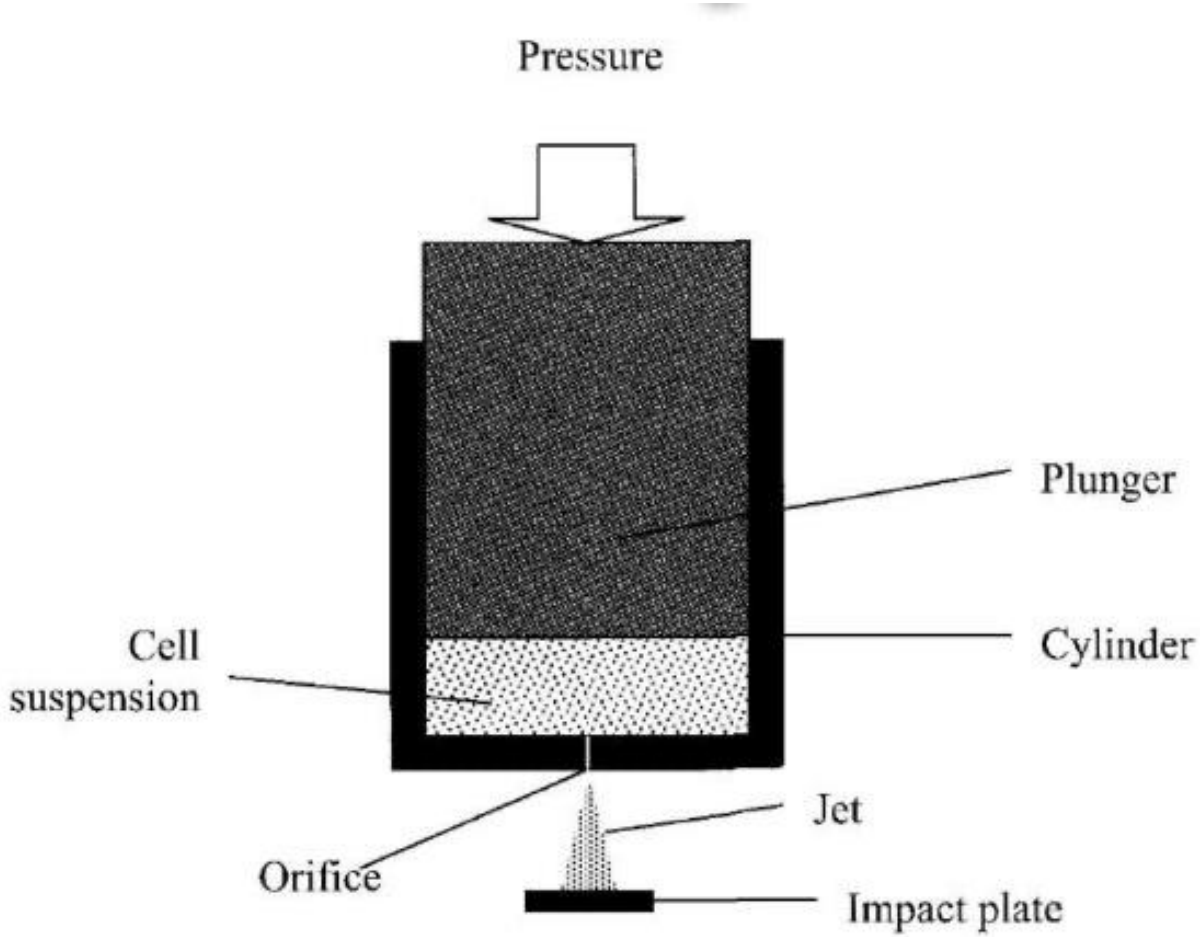
α = exponent that depends on cell toughness and can range from ~0.9 to ~2.9



Lab scale, *Constant Systems* cell disrupter

Mechanical methods: French press

- Simpler approach compared to homogenizer, suitable for **lab scale**
- High pressure applied and cells forced through a **small hole** at high speed
- Cells hit an impact plate at high speed
- Extraction efficiency described by the **same equation as high-pressure homogenizer**



Example: high-pressure homogeniser

A production strain of *E. coli* is disrupted at 5°C in a Gaulin homogenizer between 200 and $550\ \mathrm{kg}_f\ \mathrm{cm}^{-2}$. Data obtained for percentage protein release (R/R_{max}) as a function of the number of passes through the homogenizer are shown below

Number of passes	Pressure drop ($kg_f\ cm^{-2}$)				
	200	300	400	500	550
	% Protein release				
1	5.0	13.5	23.3	36.0	42.0
2	9.5	23.5	40.0	58.5	66.0
3	14.0	33.5	52.5	75.0	83.7
4	18.0	43.0	66.6	82.5	88.5
5	22.0	47.5	73.0	88.5	94.5
6	26.0	55.0	79.5	91.3	–

Example: high-pressure homogeniser

a) How many passes are required to achieve 80% release at an operating pressure of $460\ \mathrm{kg}_f\ c\mathrm{m}^{-2}$?

b) Estimate the pressure required to deliver 70% protein recovery in only two passes

Solution

First we recall

$$\ln\left(\frac{R_{max}}{R_{max}-R}\right) = kNp^{\alpha}$$

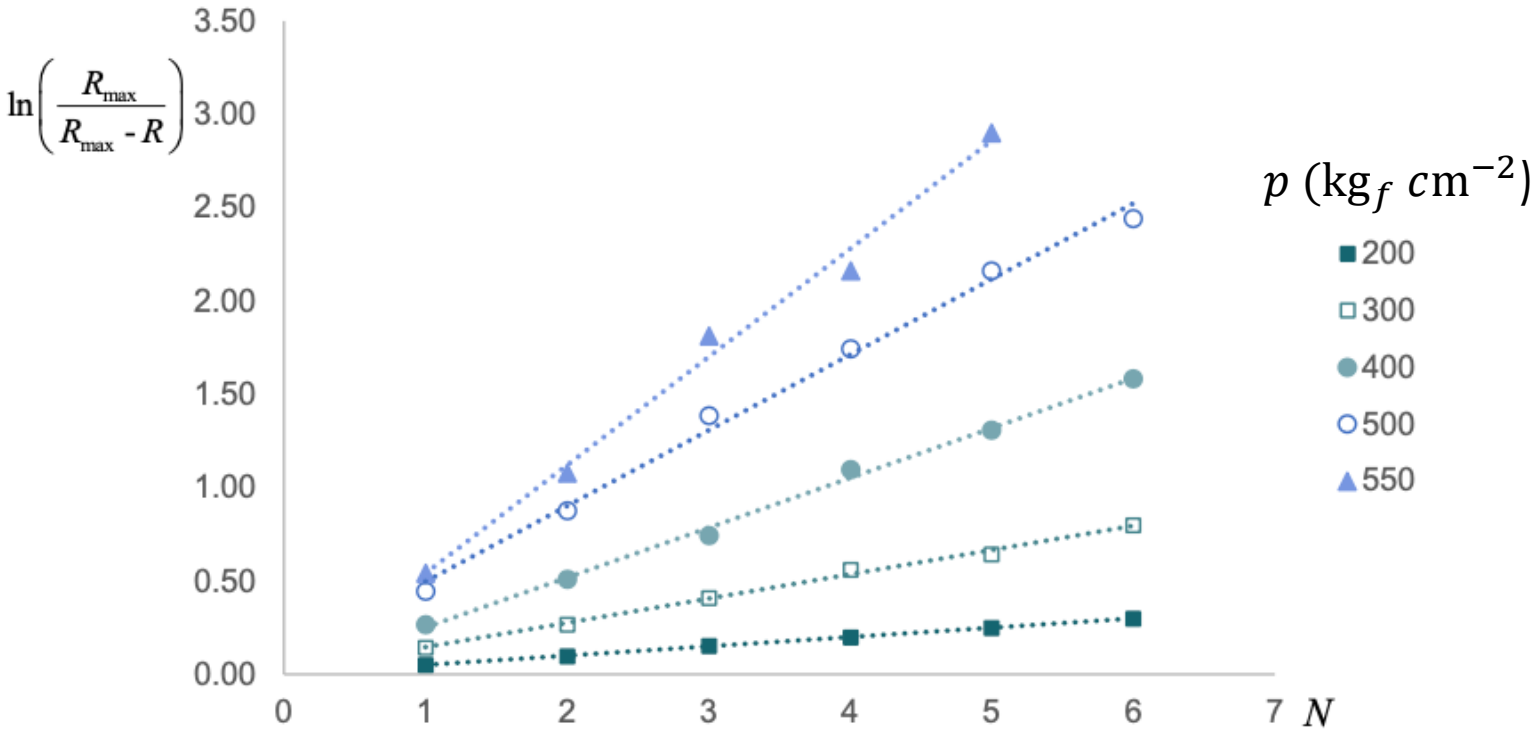
- To answer, we need to use the data provided to estimate k and α, which we can do from the data in the table.
- Table reports R/R_{max}, so we convert data to $\ln\left(\frac{R_{max}}{R_{max}-R}\right) = \ln\left(\frac{1}{1-R/R_{max}}\right)$

Solution

Number of passes	Pressure drop ($kg_f\ cm^{-2}$)				
	200	300	400	500	550
	$ln\left(\frac{R_{max}}{R_{max} - R}\right)$				
1	0.05	0.15	0.27	0.45	0.54
2	0.10	0.27	0.51	0.88	1.08
3	0.15	0.41	0.74	1.39	1.81
4	0.20	0.56	1.10	1.74	2.16
5	0.25	0.64	1.31	2.16	2.90
6	0.30	0.80	1.58	2.44	–

Solution

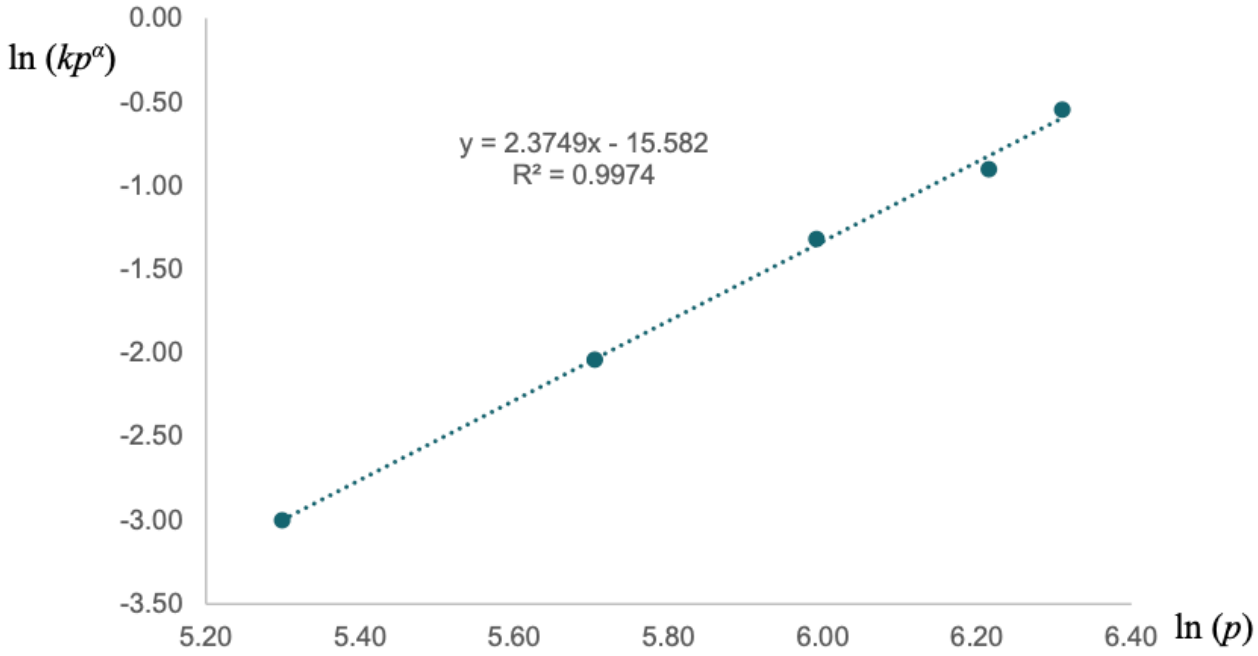
By plotting $\ln\left(\frac{R_{max}}{R_{max}-R}\right)$ vs N for datasets corresponding to different pressures we find straight lines with slope kp^{α}



Solution

From the fits in the previous slide, we obtain kp^{α} values for each value of p, shown in the table below (left). We can thus plot $\ln(kp^{\alpha})$ vs $\ln(p)$, obtaining a straight line with slope α and intercept $\ln(k)$. We obtain $\alpha = 2.37$ and $k = 1.71 \times 10^{-7}$ $\left(\mathrm{kg}_f\ \mathrm{cm}^{-2}\right)^{-\alpha}$

	Pressure drop ($\mathrm{kg}_f\ \mathrm{cm}^{-2}$)				
	200	**300**	**400**	**500**	**550**
kp^{α}	0.05	0.13	0.27	0.41	0.58



Solution: a)

How many passes are required to achieve 80% release at an operating pressure of $460\ \mathrm{kg}_f\ \mathrm{cm}^{-2}$?

$$N = \frac{1}{kp^{\alpha}} \ln\left(\frac{1}{1 - R/R_{max}}\right) = \frac{1}{1.71 \times 10^{-7} \times 460^{2.37}} \ln\left(\frac{1}{1 - 0.8}\right) = 4.6$$

Because we cannot do 4.6 passes, the answer is $N = 5$.

Solution: b)

Estimate the pressure required to deliver 70% protein recovery in only two passes.

$$p = \left(\frac{1}{Nk}\ln\left(\frac{1}{1 - R/R_{max}}\right)\right)^{1/\alpha} = \left(\frac{1}{2 \times 1.71 \times 10^{-7}}\ln\left(\frac{1}{1 - 0.7}\right)\right)^{1/2.37}$$

$$= 578 \ \mathrm{kg}_f \ \mathrm{cm}^{-2}$$